INDIA INNOVATES 2026

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Bolleans

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PROBLEM STATEMENT

Challenge:

Urban gridlock causes massive economic loss, increases carbon emissions, and critically delays emergency response vehicles, costing lives.

Existing Gaps:

- Blind Infrastructure: Traditional traffic lights use fixed timers, completely ignoring real-time congestion.
- Reactive Systems: Current emergency sensors change lights too late, leaving ambulances trapped behind already-stopped cars.
- Unsafe AI: Standard AI models change lights unpredictably, ignoring human reaction times (the "Dilemma Zone") and causing rear-end collisions.

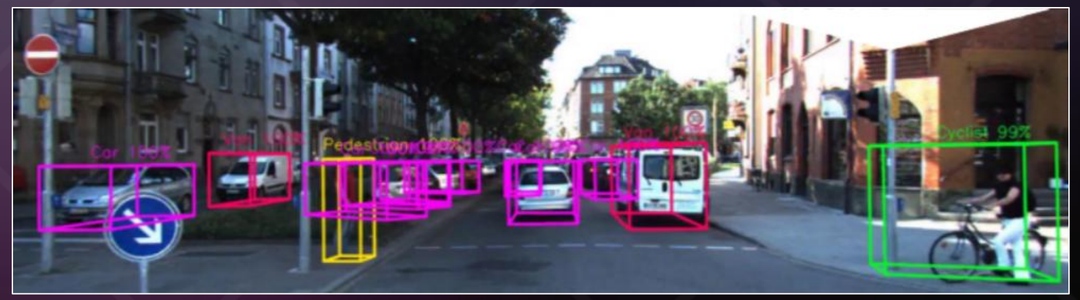
Why It Matters:

Bridge AI efficiency with physical safety to clear emergency routes early, saving lives and improving traffic flow.

SOLUTION

Proposed Approach:

- Vision: Uses real-time object detection (YOLO)
- Mathematics: Bird's-Eye View mapping to calculate live queue densities and vehicle speeds.
- Dynamic Optimization: Replaces fixed timers with Multi-Agent AI (MAPPO) that dynamically extends or ends green lights based on exact traffic demand.
- Proactive Preemption: Calculates ambulance ETA to safely flush conflicting traffic before the ambulance arrives, creating an open green corridor.



Innovation:

- Dilemma Zone Mitigation: AI outputs are bound by deterministic physics to eliminate sudden-braking accidents during dynamic light changes.

Practicality:

- Cost Effective: Uses existing CCTV cameras; no digging for underground sensors.
- Plug-and-Play: Connects directly to legacy traffic control boxes without rewiring.
- Computationally Efficient: Can run on edge devices.

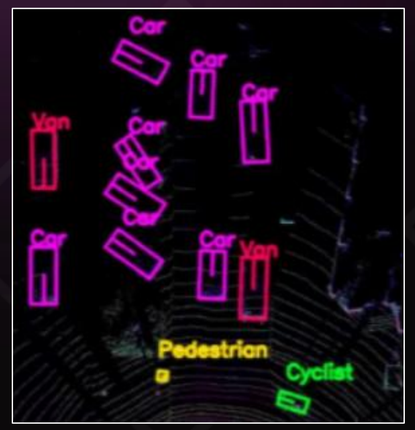


Fig:
Visual representation of Real-time YOLO object detection (top) transformed into a 3D Bird's-Eye View spatial map (bottom).

ARCHITECTURE

Phase 1: Edge Perception & Vehicle Tracking

YOLO detection outputs vehicle detections D_t per frame. Each vehicle i represented as:

$$D_{i,t} = [u_{min}, v_{min}, u_{max}, v_{max}, c, p]$$

Where (u, v) are pixel coordinates, c is class label, $p \in [0, 1]$ is confidence probability.

Centroid Extraction

Bottom-center pixel coordinate where tire touches road:

$$P_{pixel} = \left[\frac{u_{min} + u_{max}}{2}, v_{max}, 1\right]^T$$

Tracking State

Kalman Filter (ByteTrack) predicts next state, maintains unique ID by minimizing cost matrix between predicted and actual bounding boxes.

Phase 2: Geometric & Kinematic Extraction

Mapping 2D pixels to 3D physical coordinate system (Bird's Eye View).

01

Homography Transformation

Multiply pixel coordinate by calibrated 3×3 Homography matrix $\mathbf{H}$:

$$\begin{bmatrix} X' \\ Y' \\ w \end{bmatrix} = \mathbf{H} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

Normalize by w for physical coordinates in meters:

$$P_{world}(t) = \left(\frac{X'}{w}, \frac{Y'}{w}\right)$$

02

Kinematic Vectors

Calculate velocity vector using discrete time step Δt:

$$\vec{v}_t = \frac{P_{world}(t) - P_{world}(t - \Delta t)}{\Delta t}$$

03

Queue Density

For lane L_j, queue length $q_j(t)$ sums vehicles where $|\vec{v}| < v_{threshold}$ (approaching zero speed).

Phase 3: The AI Brain

MAPPO Reinforcement Learning Formulation

State Tensor S_t

Concatenated matrix of queue lengths, average approach speeds, and current active phase one-hot encoded:

$$S_t = [\mathbf{Q}_t \quad \mathbf{V}_t \quad \Phi_t]$$

Policy Network (Actor)

Processes S_t through neural network parameterized by θ to output policy $\pi_\theta(A_t|S_t)$ — probability distribution of action $A_t \in \{0, 1\}$ (0: Maintain, 1: Change).

Reward Function R_t

Trained to maximize reward that penalizes long queues and delays:

$$R_t = -\sum_{j \in Lanes} (\alpha \cdot q_{j,t} + \beta \cdot d_{j,t}^2)$$

Where α, β are weighting coefficients.

Phase 4: Deterministic Safety Logic

Timer logic implementation when AI outputs $A_t = 1$ (Change Phase).

1

Yellow Light Duration T_{yellow}

ITE kinematic formula for stopping distance:

$$T_{yellow} = t_r + \frac{v_{approach}}{2a + 2gG}$$

Where $t_r \approx 1.0\text{s}$ (reaction time), $v_{approach}$ is 85th percentile speed, $a \approx 3.0\ \text{m/s}^2$ (deceleration), g is gravity, G is road grade.

2

All-Red Clearance $T_{clearance}$

Based on intersection geometry:

$$T_{clearance} = \frac{W_{intersection} + L_{vehicle}}{v_{approach}}$$

Where W is intersection width in meters, L is standard vehicle length.

3

Countdown Timer Function

Road B calculates total wait time upon Road A initiating T_{yellow}:

$$T_{total_wait} = T_{yellow} + T_{clearance}$$

Smart display updates iteratively:

$$Timer_{display}(t) = \lceil T_{total_wait} - t_{elapsed} \rceil$$

When $Timer_{display}(t) = 0$, Road A's clearance completes and Road B's phase Φ_t sets to Green.

Phase 5: Emergency Vehicle Preemption

Continuous vector projection triggers hardware interrupt for emergency vehicles.

Condition Check

If c_{label} = 'Emergency_Vehicle' and $p > \tau_{confidence}$ (threshold), initiate tracking protocol.

ETA Projection

Calculate Euclidean distance D from ambulance's $P_{world}(t)$ to lane stop line coordinates:

$$ETA_{emergency} = \frac{D_{stop_line}}{|\vec{v}_{emergency}|}$$

Preemption Trigger

System continuously evaluates inequality:

$$ETA_{emergency} \leq (T_{yellow} + T_{clearance} + \epsilon)$$

If true, fires hardware interrupt. ϵ is safety buffer (e.g., 2 seconds) ensuring intersection flushes before ambulance arrival.

TECHNOLOGY USED

- YOLOv10 & ByteTrack(Perception): Enables real-time, ultra-low latency vehicle counting and precise emergency vehicle classification.

- PyTorch / MAPPO (AI Engine): Powers the Multi-Agent Reinforcement Learning, allowing the system to scale efficiently across multiple intersections without centralized lag.

- OpenCV & NumPy (Geometric Processing): Executes the homography matrix math to instantly convert 2D camera pixels into 3D physical world distances.

- SUMO & TraCI (Digital Twin / Simulation): Provides a microscopic testing environment to rigorously validate the AI and kinematic safety logic before real-world deployment.

FEATURE / USP

- Kinematic Safety Net (Technical Edge): Unlike pure AI that causes sudden-braking accidents, our AI is bound by deterministic physics formulas to guarantee safe, crash-proof phase transitions.

- Proactive Emergency Corridors (Innovation): Uses continuous ETA vector math to flush intersections before an ambulance arrives, replacing flawed, reactive sensors.

- Adaptive Smart Timers (User Experience): Dynamically calculates and displays red-light countdowns only during deterministic transition phases, eliminating driver anxiety without restricting AI flexibility.

- 3D BEV Perception (Technical Edge): Uses homography matrices to convert standard 2D CCTV pixels into precise 3D real-world queue metrics, avoiding expensive LIDAR or road sensors.

REFERENCES / LINKS

- Yu, C., et al. (2022). “The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games.”
- Wang, A., et al. (2024). “YOLOv10: Real-Time End-to-End Object Detection.”
- Zhang, Y., et al. (2022). “ByteTrack: Multi-Object Tracking by Associating Every Detection Box.”
- Zhou, R., et al. (2024). “Multimodal Traffic Signal Control via Constrained Deep Reinforcement Learning.”
- Chen, Y., et al. (2024). “An emergency vehicle traffic signal preemption system considering queue spillbacks along routes and negative impacts on non-priority traffic.”
- Zhou, Z., et al. (2025). “A deep reinforcement learning model for large-scale traffic signal control based on graph meta-learning using local subgraphs.”
- Kinematic Safety: Institute of Transportation Engineers (ITE) Traffic Engineering Handbook.
- Eclipse SUMO & TraCI API:
- OpenCV Homography Reference Documentation

THANK YOU